

Formal Grammar

CSG, CFG, Parse Tree, Ambiguity, Non-Determinism

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Lecture #1 out of 10
80 minutes

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Notation

Chomsky Hierarchy

Parse Tree

Ambiguity

Non-determinism



I promise, there will be no more **formalism**
than it's necessary!

Chapter #1: Notation

By the way, if a language is simple, it's possible to do it without a grammar, for example (we just split the text by a space):

```
1 | PRINT 42
2 | PRINT 256
3 | PRINT 0
```

A practical example: I have a project **Xembly**, which is using ANTLR4 for parsing its own language:

```
1 | XPATH "/car/price";  
2 | SET "$2000";  
3 | ATTR "time", "2023/02/01";
```

However, I have a task in the backlog: get rid of the grammar and use string manipulations instead, because it's faster.

A **grammar** is a finite set of formal rules for generating (!) syntactically correct sentences. Pay attention to the word “formal.” A grammar may be informal, if the rules are informal. For example:

“Commands go one after another sometimes with arguments”

This is a rule, but it is not formal and may not be understood by a computer.

Assume, we want to create a new programming language (very similar to Basic), which will allow us to write programs that look like this:

```
1 | 10 PRINT "What is your name?"  
2 | 20 INPUT X  
3 | 30 PRINT "Hello,", X
```

It's impossible (or very hard) to parse this program by splitting strings, for example, because of the possible commas inside the "Hello," string.

A formal **grammar** G , according to Noam Chomsky (1956), is a tuple $\langle N, T, P, S \rangle$, where:

- $N = \{P_{\text{rogram}}, L_{\text{ine}}, N_{\text{umber}}, C_{\text{ommand}}, A_{\text{rgument}}, \dots\}$ (**non-terminals** or **variables**)
- $T = \{10, 20, \text{PRINT}, X, ,, \text{"Hello"}, \dots\}$ (**terminals** or **alphabet**)
- $P = \{\dots\}$ (**production rules**)
- $S \in N$ (**start symbol**)

By the way, $N \cap T = \emptyset$.

A **language** that can be built by G is denoted as $\mathbf{L}(G)$: set of all strings that can be generated by G .

A **production rule** specifies a replacement of its **left-hand side** with its **right-hand side**, for example:

$$1. L_{\text{ine}} \rightarrow N_{\text{umber}} \text{ INPUT } A_{\text{rgument}}$$

$$2. N_{\text{umber}} \rightarrow 10$$

$$3. N_{\text{umber}} \rightarrow 20$$

Formally, a production rule is (using **Kleene star**, by Stephen Kleene):

$$\begin{aligned} (T \cup N)^* n (T \cup N)^* &\rightarrow (T \cup N)^* \quad n \in N \\ V^* n V^* &\rightarrow V^* \quad V = (T \cup N) \end{aligned}$$

Each left-hand side must contain at least one non-terminal symbol.

Grammars are said to be **equivalent** if they produce the same language.

Chapter #2:

Chomsky Hierarchy

There are four types in Chomsky Hierarchy of grammars:

Type-0: Unrestricted grammars

Type-1: Context-sensitive grammars

Type-2: Context-free grammars

Type-3: Regular grammars

Type-0: Unrestricted Grammar

The only restriction is that α is not empty (not ϵ) in each rule:

$$\alpha \rightarrow \beta \quad \alpha, \beta \in N \cup T$$

For every unrestricted grammar G there exists some Turing machine capable of recognizing $L(G)$ and vice versa.

[[Unrestricted](#) CSG CFG Regular]

The decision problem of whether a given string s can be generated by a given unrestricted grammar is equivalent to the problem of whether it can be accepted by the **Turing machine** equivalent to the grammar. The latter problem is called the **Halting problem** and is undecidable.

Type-1: Context-Sensitive Grammar

A **context-sensitive grammar** (CSG) are “non-erasing” grammars. A grammar is **noncontracting** (or **monotonic**) if all of its production rules are of the form $\alpha \rightarrow \beta$ where the length of α is less than or equal to that of β .

Some textbooks define CSGs as non-contracting, although this is not how Noam Chomsky defined them in 1959.

A canonical example is $\{a^n b^n c^n : n \geq 1\}$.

[Unrestricted CSG [CFG](#) Regular]

Type-2: Context Free Grammar

A **context-free grammar** (CFG) is a grammar in which the left-hand side of each production rule consists of only a single non-terminal symbol, for example:

$$p_1: P_{\text{rogram}} \rightarrow P_{\text{rogram}} L_{\text{ine}}$$

$$p_2: P_{\text{rogram}} \rightarrow \epsilon$$

$$p_3: L_{\text{ine}} \rightarrow I_{\text{nteger}} C_{\text{ommand}} T_{\text{ail}}$$

$$p_4: T_{\text{ail}} \rightarrow T_{\text{ail}} A_{\text{rgument}}$$

$$p_5: T_{\text{ail}} \rightarrow \epsilon$$

$$p_6: I_{\text{nteger}} \rightarrow 10$$

$$p_7: I_{\text{nteger}} \rightarrow 20$$

[Unrestricted CSG [CFG](#) Regular]

Derivation process may be described using $\Rightarrow_{p_i}$ notation:

$$\begin{aligned}
 P &\Rightarrow_{p_1} \mathbf{P L} \\
 &\Rightarrow_{p_3} P \mathbf{I C T} \\
 &\Rightarrow_{p_8} P \mathbf{30 C T} \\
 &\Rightarrow_{p?} P \mathbf{30 PRINT T} \\
 &\Rightarrow_{p_1} \mathbf{P L 30 PRINT T} \\
 &\Rightarrow_{p?} \dots
 \end{aligned}$$

[Unrestricted CSG [CFG](#) Regular]

We can say that “ G derives in zero or more steps”: $\xRightarrow{*}_G$ (it is **reflexive transitive closure** of $\Rightarrow_G$). For example:

$$P \xRightarrow{*}_G P \ L \ 30 \ \text{PRINT} \ T$$

[Unrestricted CSG [CFG](#) Regular]

Languages generated by context-free grammars are known as **context-free languages** (CFL).

Not all languages can be generated by CFGs.

[Unrestricted CSG [CFG](#) Regular]

The **language equality** question (do two given context-free grammars generate the same language?) is undecidable.

The **language inclusion** question is also undecidable: Given two CFGs, can the first one generate all strings that the second one can generate?

[Unrestricted CSG [CFG](#) Regular]

The **emptiness problem** (whether the grammar generates any terminal strings at all), is undecidable for context-sensitive grammars, but decidable for CFGs.

[Unrestricted CSG [CFG](#) Regular]

Leftmost derivation: always expands leftmost non-terminal.

There are **left recursive** CFGs: when non-terminals stay always on the left side of the right-side hand of the rule. Similarly, there are **right recursive** CFGs.

[Unrestricted CSG CFG [Regular](#)]

Type-3: Regular Grammar

In a **regular grammar** all production rules have at most one non-terminal symbol in the rightmost or leftmost position in the rule (A and B are non-terminals and a is a string of terminals):

$$A \rightarrow a$$

$$A \rightarrow a B \quad (\text{right-linear grammar})$$

$$A \rightarrow B a \quad (\text{left-linear grammar})$$

$$A \rightarrow \epsilon$$

[Unrestricted CSG CFG [Regular](#)]

Left-linear grammar is just another name for left-regular grammar (the same for right-).

[Unrestricted CSG CFG [Regular](#)]

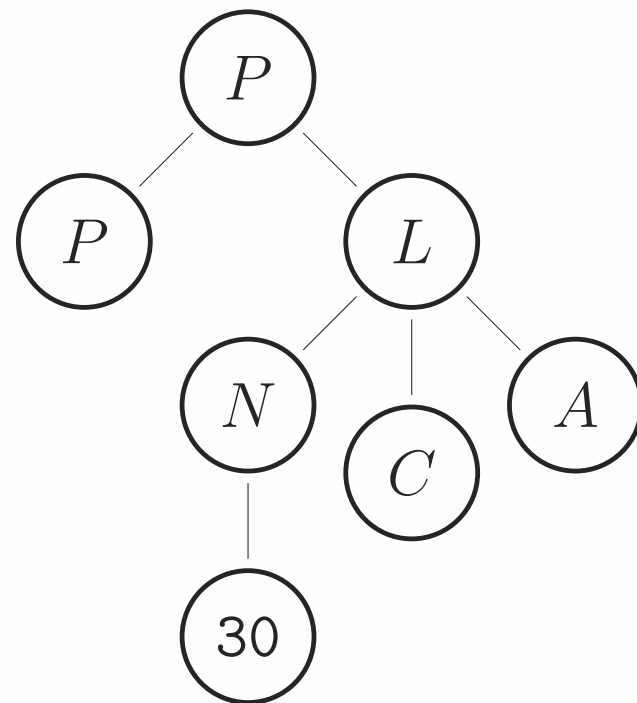
Some textbooks and articles disallow empty rules (with ϵ).

[Unrestricted CSG CFG [Regular](#)]

A regular grammar generates exactly the language a nondeterministic finite automaton accepts.

Chapter #3: Parse Tree

A **parse tree** (parsing tree, derivation tree, concrete syntax tree) is an ordered, rooted tree that represents the syntactic structure of a string according to some CFG.

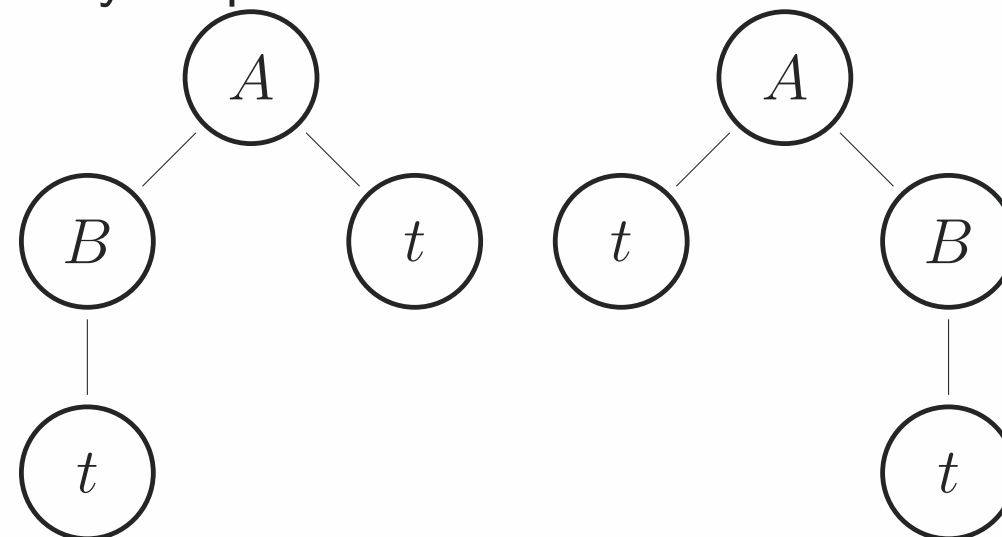


Chapter #4: Ambiguity

An **ambiguous grammar** is a CFG for which there exists a string that can have more than one leftmost derivation or parse tree. For example, this grammar:

$$\begin{aligned} A &\rightarrow B t \mid t B \\ B &\rightarrow t \end{aligned}$$

May be parsed as two different trees:



Chapter #5:

Non-determinism

Non-deterministic CFG:

$$A \rightarrow B x$$

$$A \rightarrow B y$$

$$A \rightarrow B z$$

Backtracking in a parser is required in order to parse this grammar.

By using **left factoring** it is possible to remove non-determinism:

$$A \rightarrow B C$$

$$C \rightarrow x \mid y \mid z$$

